Roll No.: _____

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – December 2025

Third Semester

Artificial Intelligence & Data Science (Medical Engineering)

23AID204 Advanced Data Structures & Algorithm Analysis

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Develop skills for analyzing algorithmic strategies
CO02	Analyse and apply appropriate algorithmic technique for a given problem
CO03	Implementing standard algorithms on arrays, strings, trees and graphs
CO04	Visualize multidimensional geometry of data structure and concurrency.

Answer all questions

1) Explain the height, depth, degree, and level of nodes in binary trees with a suitable diagram or example. [10][CO01][BTL 1]
2) Construct a Binary Search Tree from the sequence [45, 20, 60, 10, 30, 50, 70, 25]. Show the inorder, preorder, and postorder traversals. [10][CO01][BTL 3]
3) Write the pseudocode for the insertion and deletion of a max heap binary tree. [10][CO01][BTL 2]
4) Consider the array [40, 15, 20, 10, 8, 12, 5].
   (a) Build a Max Heap from this array.
   (b) Perform Heap Sort and show intermediate heap states. [10][CO02][BTL 4]
5) Insert the elements [30, 20, 40, 10, 25, 35, 50, 5] into an AVL Tree.
   Show the tree after each insertion and mention which rotation was applied. [10][CO02][BTL 5]

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**10,4,10**	1,2,3	**24**
CO02	**6,10,10**	2,4,5	**26**